SCOPE OF WORK

USE THE NEW 200A, 120/208V, 3-PHASE, 4 WIRE ELECTRICAL SERVICE. USE NEW ELECTRICAL METER AND 200A DISCONNECT SWITCH FROM FEED FROM UTILITY AND PROVIDE (1) 200 A(MCB) & (1) 100 A(MLO), 120/208V, 3-PHASE ELECTRICAL PANEL "PA" & "PB" FOR THE PROPOSED TENANT SPACE. ALL NECESSARY EQUIPMENT, WIRING AND LIGHTING FOR THE PROJECT SPACE. COORDINATE WITH G.C FOR LOW VOLTAGE WIRING.

ELECTRICAL PLAN NOTES

1. ELECTRICAL CONTRACTOR SHALL REVIEW ALL DRAWINGS OF THIS SET.
2. CONTRACTOR TO VERIFY THAT ALL EQUIPMENT SHOWN AS EXISTING MATCHES THE DESCRIPTIONS AND SPECIFICATIONS SHOWN ON DRAWINGS AND SCHEDULES. IF DIFFERENT, NOTIFY ARCHITECT/ENGINEER BEFORE BIDDING, ORDERING, OR PROCEEDING WITH WORK.
3. ELECTRICAL CONTRACTOR SHALL SUPPLY AND INSTALL ALL NEW ELECTRICAL WORK INDICATED. CONSTRUCTION SHALL BE IN ACCORDANCE WITH DRAWINGS AND APPLICABLE SPECIFICATIONS. IF A PROBLEM IS ENCOUNTERED IN COMPLYING WITH THIS REQUIREMENT, CONTRACTOR SHALL NOTIFY THE OWNER OR HIS REPRESENTATIVE AS SOON AS POSSIBLE AFTER DISCOVERY OF THE PROBLEM AND SHALL NOT PROCEED WITH THAT PORTION OF THE WORK UNTIL OWNER HAS DIRECTED CORRECTIVE ACTION TO BE TAKEN.
4. ELECTRICAL CONTRACTOR SHALL VISIT JOB SITE AND FAMILIARIZE HIMSELF WITH ALL CONDITIONS AFFECTING ELECTRICAL AND COMMUNICATIONS INSTALLATION AND MAKE PROVISIONS AS TO THE COST THEREOF. EXISTING CONDITIONS OF ELECTRICAL EQUIPMENT, LIGHT FIXTURES, ETC... THAT ARE PART OF THE FINAL SYSTEM SHALL BE VERIFIED BY THE CONTRACTOR PRIOR TO SUBMITTING HIS BID.
5. ALL WORK SHALL BE PERFORMED IN ACCORDANCE WITH THE 2020 EDITION OF THE NATIONAL ELECTRIC CODE AND ALL CODES AND ORDINANCES OF THE AUTHORITY HAVING JURISDICTION.
6. DO NOT SCALE THE ELECTRICAL DRAWINGS. REFER TO ARCHITECTURAL PLANS AND ELEVATIONS FOR EXACT LOCATION FOR ALL EQUIPMENT. CONFIRM WITH OWNER'S REPRESENTATIVE.
7. ALL ELECTRICAL NOT BEING REUSED MUST BE REMOVED IN ITS ENTIRETY.
8. ALL CONDUIT IN OR UNDERGROUND OR IN CONCRETE MUST BE RIGID GALVANIZED STEEL.
9. CIRCUIT BREAKERS AND PANELS TO BE BOLT ON TYPE.
10. ALL EQUIPMENT SHALL BE APPROVED BY UL OR OTHER NATIONALLY RECOGNIZED TESTING COMPANY.
11. ALL RECEPTACLES SHALL BE GROUNDED AS REQUIRED BY NEC 250.146
12. SUBMIT SERVICE ENTRANCE EQUIPMENT FOR SEPARATE APPROVAL.
13. ALL LOW VOLTAGE MUST BE IN CONDUIT TO ABOVE THE DROP CEILING. BRIDAL RINGS OR "J" HOOKS REQUIRED.
14. SEPARATE PERMITS ARE REQUIRED FOR ALL LOW VOLTAGE SUCH AS TELEPHONE, DATA, THERMOSTAT, MUSIC, ALARMS ETC.
15. SEPARATE PERMIT REQUIRED FOR SIGNAGE.
16. PRIOR TO ANY CONSTRUCTION WORK BEGINNING AN ON-SITE MEETING WITH GENERAL CONTRACTORS IS REQUIRED.
17. ELECTRICIAN MUST BE ON SITE FOR ALL INSPECTIONS.
18. MINIMUM WIRE SIZE SHALL BE #12 A.W.G. EXCLUDING CONTROL WIRING. ALL CONDUCTORS SHALL BE COPPER AND UNLESS OTHERWISE NOTED THHN INSULATION.
19. OUTLET BOXES SHALL BE PRESSED STEEL IN DRY LOCATIONS, PLASTIC AND CAST ALLOY WITH THREADED HUBS IN WET OR DAMP LOCATIONS, AND SPECIAL ENCLOSURES FOR OTHER CLASSIFIED AREAS.
20. IT IS NOT THE INTENT OF THESE PLANS TO SHOW EVERY MINOR DETAIL OF CONSTRUCTION. THE CONTRACTOR IS EXPECTED TO FURNISH AND INSTALL ALL ITEMS FOR A COMPLETE ELECTRICAL SYSTEM AND PROVIDE ALL REQUIREMENTS NECESSARY FOR EQUIPMENT TO BE PLACED IN PROPER WORKING ORDER.
21. ELECTRICAL SYSTEM SHALL BE COMPLETE AND EFFECTIVELY GROUNDED AS REQUIRED BY THE N.E.C. OR LOCAL CODES.
22. ALL MATERIALS SHALL BE NEW AND BEAR UNDERWRITERS' LABELS WHERE APPLICABLE.
23. ALL WORK SHALL BE PERFORMED BY A LICENSED ELECTRICAL CONTRACTOR IN A FIRST CLASS WORKMANLIKE MANNER. THE COMPLETED SYSTEM SHALL BE FULLY OPERATIVE AND ACCEPTED BY ENGINEER/ARCHITECT.
24. ALL WORK SHALL BE COORDINATED WITH OTHER TRADES TO AVOID INTERFERENCE WITH THE PROGRESS OF CONSTRUCTION.
25. ELECTRICAL CONTRACTOR SHALL GUARANTEE ALL MATERIALS AND WORKMANSHIP FREE FROM DEFECTS FOR A PERIOD OF NOT LESS THAN ONE YEAR FROM DATE THAT CERTIFICATE OF OCCUPANCY IS ISSUED. WARRANTY SHALL BE PROVIDED IN WRITING. PROVIDE COPY TO LL.
26. CORRECTION OF ANY DEFECTS SHALL BE COMPLETED WITHOUT ADDITIONAL CHARGE AND SHALL INCLUDE REPLACEMENT OR REPAIR OF ANY OTHER PHASE OF THE INSTALLATION WHICH MAY HAVE BEEN DAMAGED THEREBY.
27. ALL REQUIRED INSURANCE SHALL BE PROVIDED FOR PROTECTION AGAINST PUBLIC LIABILITY AND PROPERTY DAMAGE FOR THE DURATION OF THE WORK.
28. CONTRACTOR SHALL PAY FOR ALL PERMITS, FEES, INSPECTIONS AND TESTING. CONTRACTOR TO OBTAIN PERMIT AND APPROVED SUBMITTALS PRIOR TO BEGINNING WORK OR ORDERING EQUIPMENT.
29. THE ELECTRICAL INSTALLATION SHALL MEET ALL STANDARD REQUIREMENTS OF POWER AND TELEPHONE COMPANIES.
30. CONTRACTOR SHALL COORDINATE WITH MECHANICAL DRAWINGS AND PROVIDE ALL NECESSARY CONTROL WIRING.
31. ALL CIRCUIT BREAKERS FEEDING MECHANICAL EQUIPMENT SHALL BE HACR TYPE CIRCUIT BREAKERS.
32. PROVIDE AND INSTALL CONDUIT, CONDUCTORS, PULL WIRES, BOXES, COVER PLATES, DEVICES, ETC. FOR ALL OUTLETS AS INDICATED.
33. MATERIALS, PRODUCTS, AND EQUIPMENT, INCLUDING ALL COMPONENTS THEREOF, SHALL BE NEW AND SUCH AS APPEAR ON THE UL LIST OF APPROVED ITEMS AND SHALL MEET OR EXCEED THE REQUIREMENTS OF N.E.C. NEMA, AND IECE.
34. CONTRACTOR SHALL SUBMIT SHOP DRAWINGS OR CUT SHEETS OF LIGHTING FIXTURES, SWITCHES, AND OTHER ELECTRICAL ITEMS FOR APPROVAL BY ENGINEER/ARCHITECT.
35. ELECTRICAL CONTRACTOR SHALL BE RESPONSIBLE FOR ALL CUTTING, PATCHING AND FIRED CAULKING REQUIRED OF HIS WORK.
36. ELECTRICAL CONTRACTOR SHALL LABEL ALL PANELS W/TYPE WRITTEN DIRECTORIES.
37. ALL ELECTRICAL AND COMMUNICATIONS OUTLETS TO BE AT 24" A.F.F. UNLESS NOTED OTHERWISE, AND VERTICALLY MOUNTED.
38. ALL LIGHT SWITCHES TO BE AT 42" A.F.F.
39. ALL ELECTRICAL WIRING SHALL BE THE RESPONSIBILITY OF THE ELECTRICAL CONTRACTOR. ALL ELECTRICAL WIRING FOR HVAC SYSTEM INCLUDING CONTROLS, THERMOSTATS, POWER, ETC. SHALL BE THE RESPONSIBILITY OF THE ELECTRICAL CONTRACTOR.
40. BREAKER AND PANELS -- ALL CURRENT CARRYING BUSSES SHALL BE COPPER. ALL GROUND BUS BARS SHALL BE COPPER. PANEL BOARD ENCLOSURES SHALL BE FURNISHED WITHOUT PRE-PUNCHED CONCENTRIC HOLES. A.I.C. RATINGS SHALL BE AS INDICATED ON PANEL BOARD SCHEDULES.
41. DISCONNECT SWITCHES SHALL BE H.P. RATED, GENERAL DUTY, QUICK-MAKE, QUICK-BREAK ENCLOSURES AS REQUIRED BY EXPOSURE.
42. MOTOR STARTERS SHALL BE MANUAL OR MAGNETIC, WITH OVERLOAD RELAYS IN EACH HOT LEG.
43. THE TERM "PROVIDE" USED IN THE CONSTRUCTION DOCUMENTS AND SPECIFICATIONS INDICATES THE CONTRACT SHALL FURNISH AND INSTALL.
44. CONTRACTOR SHALL CONFIRM WITH ANY AND ALL REQUIREMENTS SUCH AS: LUG SIZE RESTRICTIONS, CONDUIT ENTRY, TRANSFORMER SIZE, SCHEDULED DOWN TIME FOR OWNERS CONFIRMATION, ETC. ANY CONFLICTS SHALL BE BROUGHT TO ATTENTION OF THE ENGINEER BEFORE PROCEEDING WITH ANY WORK.
45. VOLTAGE DROP FOR ALL BRANCH CONDUCTORS SHALL NOT EXCEED 3%. WHERE VOLTAGE DROP EXCEEDS 3%, CONTRACTOR SHALL INCREASE SIZE OF CONDUCTORS.
46. CONTRACTOR SHALL PROVIDE GFI TYPE BREAKER FOR ALL EXTERIOR 120V CIRCUITS OR GFI PROTECTION -- FOR THE WHOLE CIRCUIT.
47. GAS PIPING SHALL BE BONDED.
48. ELECTRICAL CONTRACTOR SHALL COORDINATE SERVICE ENTRY WITH SERVICE PROVIDER PRIOR TO DETERMINING EXACT LOCATION OF THE METER BOX IN ORDER TO AVOID DISCREPANCIES BETWEEN DRAWINGS AND JOB CONDITIONS.
49. ALL OUTDOOR EQUIPMENT SHALL BE WEATHERPROOF.
50. CONSTRUCTION "AS BUILT" DRAWINGS AND DOCUMENTS SHALL BE PROVIDED TO THE OWNER WITHIN 30 DAYS AFTER THE DATE OF ACCEPTANCE. PROVIDE A COPY TO LL.
51. OPERATION MANUALS AND MAINTENANCE MANUALS SHALL BE PROVIDED TO THE BUILDING OWNER.
52. ABSOLUTELY NO FLEXIBLE CONDUIT IS PERMITTED IN DEMISING WALLS. FLEXIBLE CONDUIT IS PERMITTED FOR SHORT FINAL CONNECTIONS ONLY (6'-0" OR LESS).
53. EXPOSED CONDUIT SHALL BE INSTALLED IN STRAIGHT LINES, PARALLEL OR IN RIGHT ANGLES TO THE BUIDING STRUCTURE. DO NOT LOOP EXCESS FLEXIBLE CONDUIT IN CEILING SPACE OR WALL CAVITY. NO CONDUIT TO BE SUPPORTED FROM THE ROOF DECK.
54. CABLE TYPES AC AND NM CABLES ARE NOT ACCEPTABLE. TYPE MC CABLE, ELECTRIC METALLIC TUBING (EMT) AND RIGID GALVANIZED CONDUIT ARE PERMITTED.
55. ALL EQUIPMENT, DEVICES AND FIXTURES SHALL BE GROUNDED IN COMPLIANCE WITH NEC AND UL REQUIREMENTS.
56. ALL PANELS TO BE UL LABELED WITH BOLT-ON TYPE CIRCUIT BREAKERS.
57. 7-DAY 24-HOUR TIME CLOCK IS REQUIRED TO CONTROL STOREFRONT ENTRY LIGHTS, SHOW WINDOW LIGHTS, SHOW WINDOW RECEPTACLES AND STOREFRONT SIGNAGE. ILLUMINATED STOREFRONT SIGNS MUST REMAIN LIT DURING ALL MALL BUSINESS HOURS.
58. TENANT IS REQUIRED TO MAKE A FIELD SURVEY OF THE EXISTING ELECTRICAL SERVICE TO ENSURE THAT THE TOTAL CONNECTED LOAD DOES NOT EXCEED THE ELECTRIC SERVICE. ANY/ALL MODIFICATIONS OR UPGRADES NEEDED ARE SUBJECT TO LANDLORD'S PRIOR APPROVAL AND WILL BE COMPLETED BY TENANT/TENANT'S GC AT TENANT'S SOLE EXPENSE.
59. ALL ELECTRICAL PANELS TO BE MOUNTED ON PLYWOOD BACKER BOARD.
60. PANEL PHASE LOADS TO BE BALANCED WITHIN 10%.

ELECTRICAL LEGEND

- JUNCTION BOX. CEILING MOUNTED / WALL MOUNTED
- CONDUIT HOMERUN TO ELECTRIC PANEL. NOTATIONS INDICATE PANEL AND CIRCUIT NUMBER. QUANTITY OF HALF ARROWS INDICATE NUMBER OF CIRCUITS UNLESS OTHERWISE INDICATED.
- WALL MOUNTED LIGHT SWITCH, LUTRON MAESTRO WIRELESS OR OTHER COMPATIBLE WITH OCCUPANCY SENSOR. 'o' INDICATES ZONE.
- WALL MOUNTED 3-WAY LIGHT SWITCH, 20A RATED
- SURFACE MOUNTED LIGHTING OR RECEPTACLE PANEL.
- METER/ CT CABINET
- DISCONNECT SWITCH
- FUSED DISCONNECT SWITCH
- JUNCTION BOX AND THERMAL DISCONNECT SWITCH
- GENERATOR
- MOTOR AND DISCONNECT SWITCH. REFER TO MECHANICAL DRAWINGS FOR HP RATINGS. NUMERICAL INDICATES AMPERAGE RATING.
- GROUND CONNECTION
- PHOTOVOLTAIC CELL FOR EXTERIOR LIGHTING
- PLAN NOTE NOTATION
- DUPLEX RECEPTACLE, 20A, 120V
- DUPLEX RECEPTACLE WITH GFCI PROTECTION, 20A, 120V
- QUAD RECEPTACLE, 20A, 120V
- SPECIAL RECEPTACLE, VOLTAGE AND AMPERAGE BASED ON CONNECTED CIRCUIT
- DISCONNECT SWITCH - FUSED
- DISCONNECT SWITCH - UNFUSED
- EVSC — ELECTRIC VEHICLE CHARGING STATION

GENERAL LIGHTING NOTES

A. ALL EMERGENCY FIXTURES SHALL BE CONNECTED TO AN UNSWITCHED HOT CONDUCTOR

EXISTING CONTIDITONS NOTES

STOP AND READ

THE CONTRACTOR AND SUB-CONTRACTORS **SHALL NOT INITIATE ANY WORK UNTIL EXISTING FIELD CONDITIONS ARE PROPERLY VERIFIED**. THIS SHALL HOLD TRUE FOR FIRST GENERATION AND 2ND GENERATION SPACES. WHEN DEMOLITION IS REQUIRED, THAT WILL BE PERMITTED TO EXPOSE CONDITIONS. THESE VERIFICATIONS SHALL INCLUDE BUT NOT LIMITED TO: DIMENSIONS BOTH HORIZONTALLY AND VERTICAL, ELECTRICAL SERVICE /PANELS LOCATION AND VOLTS/PHASE, LOCATION/QTY OF ROOF MOUNTED HVAC EQUIPMENT, CONFIRM THAT INTERIOR HVAC HUNG UNITS HAVE PROPER SUPPORT CONNECTIONS FOR EXISTING STRUCTURE, FIRE SPRINKLER MAIN RUNS, TOILET ROOM DIMENSIONS, DOOR SWING FOR DOORS TO REMAIN AND ETC. IF NOT VERIFIED AND DISCOVERED AT A LATER TIME, THE CONTRACTOR SHALL REIMBURSE THE ARCHITECT FOR THE REDESIGN FEE. THIS DOES NOT INCLUDE HIDDEN WORK I.E. PITCH OF SANITARY LINES, ACTUAL CONDITIONS OF EXISTING HVAC EQUIPMENT, STRUCTURAL COLUMNS/BEARING WALLS OR CONDITIONS OF GREASE INTERCEPTORS AND ETC.

LIGHTING FIXTURE SCHEDULE

	TYPE	DESCRIPTION	VOLT	NUMBER OF FIXTURES	LAMP TYPE	WATTS	MOUNTING
	A	PENDANT LIGHT	120	___	LED	TBD	TBD
	B	ALIGHIERI WALL SCONE	120	___	LED	TBD	TBD
	C	CONTENDER CHANDELIER	120	___	LED	TBD	TBD
	R	RECESSED LED CAN LIGHT	120	___	LED	TBD	TBD
	G	RECESSED LED CAN LIGHT	120	___	LED	TBD	TBD
	EM-1	EMERGENCY LIGHTING	120	___	LED	TBD	TBD
DM		DIMMER LIGHT SWITCH	___	___	___	___	___
3		3-WAY LIGHT SWITCH	___	___	___	___	___
		LIGHT SWITCH	___	___	___	___	___
OS	OS	OCCUPANCY SENSOR	___	___	___	___	___
3DM		3-WAY DIMMER SWITCH	___	___	___	___	___
J		JUNCTION BOX	___	___	___	___	___
		CAT5 W/TV	___	___	___	___	___

DRAWING LIST

E-0.0	ELECTRICAL NOTES AND DRAWING LIST
E-1.0	ELECTRICAL BASEMENT LIGHTING PLAN
E-1.1	ELECTRICAL MAIN LIGHTING PLAN
E-1.2	ELECTRICAL ROOF LIGHTING PLAN
E-2.0	ELECTRICAL BASEMENT POWER PLAN
E-2.1	ELECTRICAL MAIN POWER PLAN
E-2.2	ELECTRICAL ROOF POWER PLAN
E-3.0	ELECTRICAL RISER DIAGRAM & LOAD ANALYSIS
E-4.0	ELECTRICAL PANEL SCHEDULE

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ELECTRICAL NOTES AND DRAWING LIST

DRAWING NUMBER:

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LIGHTING NOTES

1. REFER TO ARCHITECTURAL DRAWINGS FOR LIGHTING FIXTURE SCHEDULES, EXACT FIXTURE LOCATIONS, EXTEND OF DEMOLITION AND FIXTURE REMOVAL.
2. THE ELECTRICAL CONTRACTOR SHALL PROVIDE LABELING FOR ALL NEW PANELS AND SHALL RELABEL ALL EXISTING PANELS, AND CIRCUIT NUMBER.
3. CIRCUIT NUMBERS NEXT TO LIGHTING FIXTURES ARE FOR LOAD PURPOSE ONLY. CONTRACTOR SHALL WIRE IN A CONSISTENT AND LOAD BALANCED MANNER.
4. ELECTRICAL CONTRACTOR SHALL INSTALL ALL CONDUIT, WIRING, JUNCTION BOXES & FLEXIBLE FINAL CONNECTIONS AS REQUIRED TO MEET NATIONAL ELECTRICAL CODE.
5. ALL DEVICES (SWITCHES, RECEPTACLES, ETC) FINISHES TO BE COORDINATED WITH ARCHITECTS.
6. CONTRACTOR TO COORDINATE WITH ARCH DRAWINGS FOR SWITCHING AND SWITCH LOCATIONS. CONTRACTOR TO ADJUST CIRCUITING AS NEEDED TO GROUP LIGHTING TO SWITCHING SCHEME.
7. CONTRACTOR SHALL COORDINATE WITH OWNER TO VERIFY EXACT REQUIREMENTS OF ALL EQUIPMEMNTS

ELECTRICAL BASEMENT LIGHTING PLAN

SCALE: 1/4"=1'=0"

MAKMO DESIGN - DESIGN WITH A DIFFERENT APPROACH

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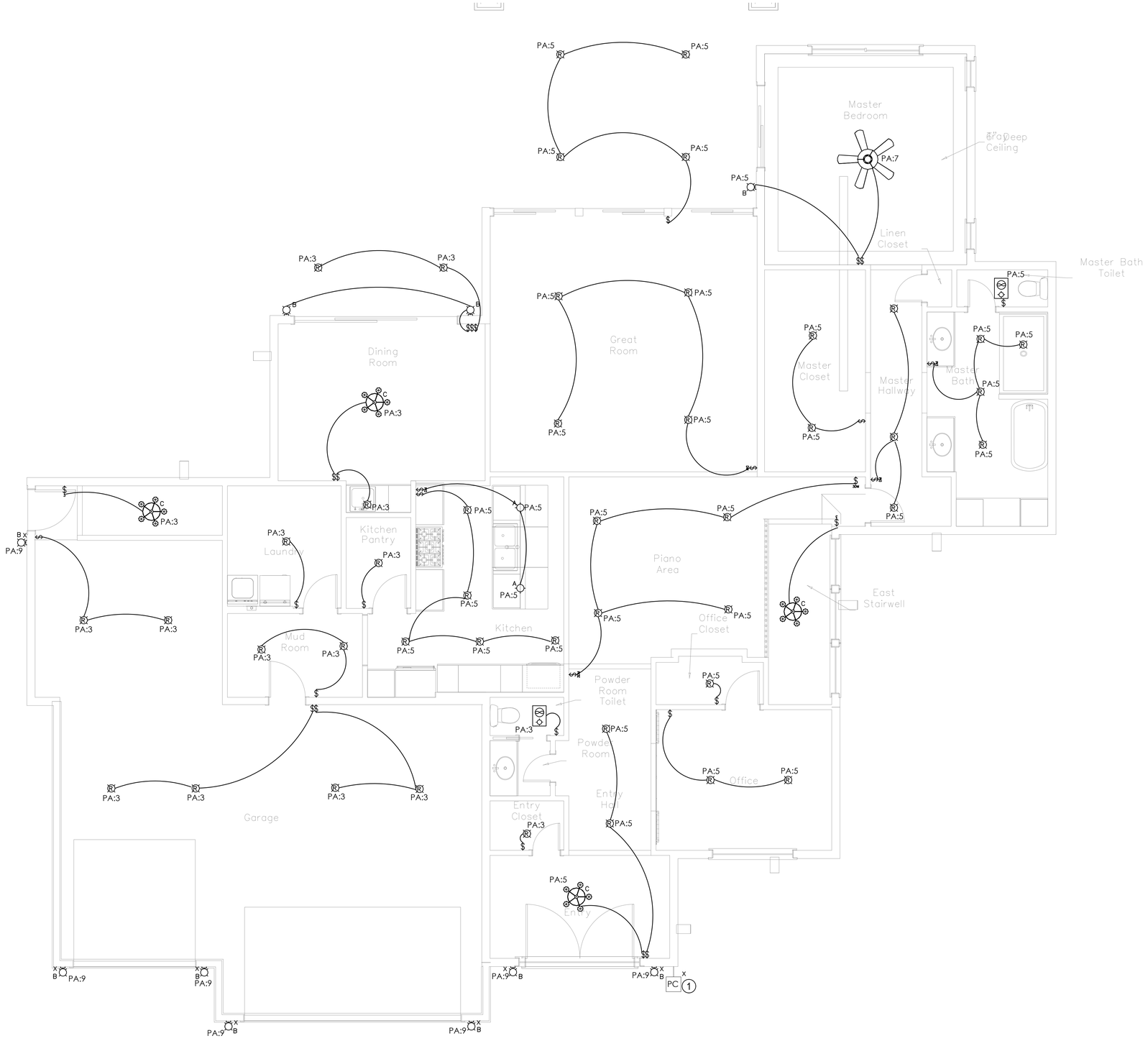
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ELECTRICAL BASEMENT LIGHTING PLAN

DRAWING NUMBER:

E-1.0



ELECTRICAL MAIN LIGHTING PLAN

SCALE: 1/4"=1'=0"

MAKMO DESIGN - DESIGN WITH A DIFFERENT APPROACH

LIGHTING NOTES

1. REFER TO ARCHITECTURAL DRAWINGS FOR LIGHTING FIXTURE SCHEDULES, EXACT FIXTURE LOCATIONS, EXTEND OF DEMOLITION AND FIXTURE REMOVAL.
2. THE ELECTRICAL CONTRACTOR SHALL PROVIDE LABELING FOR ALL NEW PANELS AND SHALL RELABEL ALL EXISTING PANELS, AND CIRCUIT NUMBER.
3. CIRCUIT NUMBERS NEXT TO LIGHTING FIXTURES ARE FOR LOAD PURPOSE ONLY. CONTRACTOR SHALL WIRE IN A CONSISTENT AND LOAD BALANCED MANNER.
4. ELECTRICAL CONTRACTOR SHALL INSTALL ALL CONDUIT, WIRING, JUNCTION BOXES & FLEXIBLE FINAL CONNECTIONS AS REQUIRED TO MEET NATIONAL ELECTRICAL CODE.
5. ALL DEVICES (SWITCHES, RECEPTACLES, ETC) FINISHES TO BE COORDINATED WITH ARCHITECTS.
6. CONTRACTOR TO COORDINATE WITH ARCH DRAWINGS FOR SWITCHING AND SWITCH LOCATIONS. CONTRACTOR TO ADJUST CIRCUITING AS NEEDED TO GROUP LIGHTING TO SWITCHING SCHEME.
7. CONTRACTOR SHALL COORDINATE WITH OWNER TO VERIFY EXACT REQUIREMENTS OF ALL EQUIPMEMNTS.

SHEET NOTES

(1) ALL EXTERIOR LIGHTS TO BE EQUIPPED WITH PHOTOCELLS. ALL HOUSE EXTERIOR LIGHTS TO BE EQUIPPED WITH PHOTOCELLS AND CONTROLLED VIA TIME CLOCK

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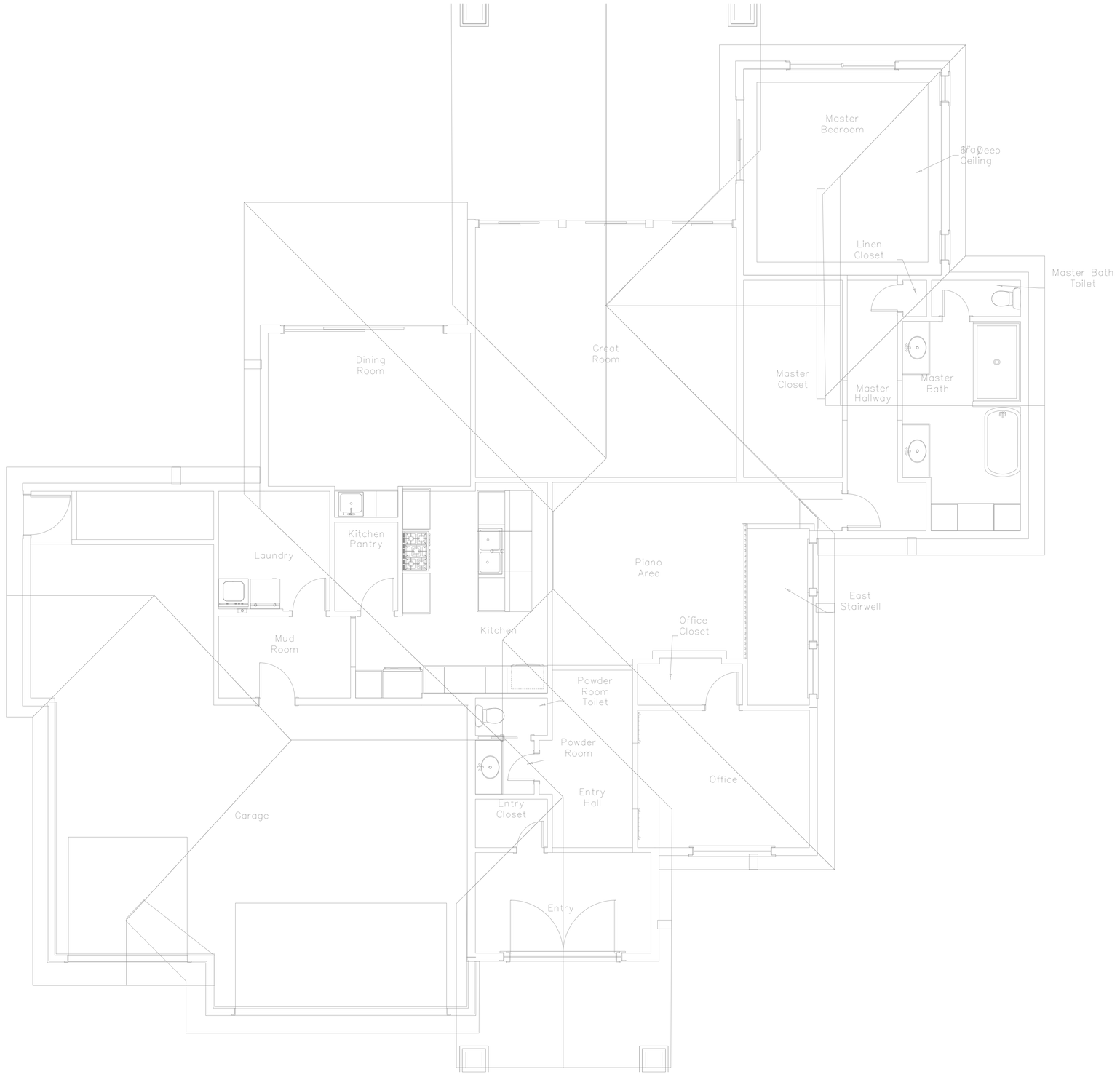
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ELECTRICAL MAIN LIGHTING PLAN

DRAWING NUMBER:

E-1.1



ELECTRICAL ROOF LIGHTING PLAN

SCALE: 1/4"=1'=0"

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ELECTRICAL ROOF LIGHTING PLAN

DRAWING NUMBER:

E-1.2



ELECTRICAL BASEMENT POWER PLAN

SCALE: 1/4"=1'=0"

MAKMO DESIGN - DESIGN WITH A DIFFERENT APPROACH

POWER NOTES

1. VERIFY EXACT REQUIREMENTS FOR ELECTRICAL EQUIPMENT WITH OWNER FOR THIS PROJECT. IN THE CASE OF A DISCREPANCY BETWEEN THE ACTUAL SELECTION OF THE EQUIPMENT AND THESE DRAWINGS, ADVISE THE OWNER BEFORE WORK
2. EXACT LOCATION OF RECEPTACLES & QUANTITY SHALL BE VERIFIED WITH ARCHITECT'S DOCUMENTS. POWER OUTLETS SHALL BE INSTALLED AT 16" A.F.F. TO BOTTOM OF OUTLET IN ALL PUBLIC AREAS. LIGHT SWITCHES SHALL BE INSTALLED AT 48" A.F.F. TO TOP OF SWITCHES IN ALL PUBLIC AREAS.
3. DRAWING IS DIAGRAMMATIC & INDICATES DIAGRAMMATIC WIRING & CIRCUIT DESIGNATIONS TO RECEPTACLES & DEVICES. FURNISH AND INSTALL ALL WIRING & CONDUITS AS REQUIRED FOR FULLY WORKABLE SYSTEM.
4. ELECTRICAL WHIPS/FLEXIBLE CONNECTIONS TO ELECTRIFIED FURNITURE SHALL BE A MAXIMUM LENGTH OF 18".
5. CONTRACTOR TO PROVIDE TYPED UPDATED PANEL DIRECTORY AS REQUIRED.
6. THE ELECTRICAL CONTRACTOR SHALL PROVIDE LABELING FOR ALL NEW PANELS AND SHALL RELABEL ALL EXISTING PANELS, AND CIRCUIT NUMBER.
7. CONTRACTOR SHALL PROVIDE AS-BUILT DRAWINGS INDICATING ACTUAL BRANCH CIRCUITING AND OUTLET LOCATION UPON COMPLETION OF WORK.
8. ELECTRICAL CONTRACTOR TO COORDINATE RECEPTACLE TYPE (NEMA TYPE) WITH EQUIPMENT (I.E. COPIER) SUPPLIER/MANUFACTURER PRIOR TO INSTALLATION.
9. PROVIDE 1¼"CONDUIT STUB-UPS AND BACK BOXES FOR ALL AV OUTLETS SHOWN. REFER TO AV PLANS FOR BOX SIZES.
10. ALL CIRCUITS TO BE TAGGED AT EACH END WITH APPROPRIATE CIRCUIT NUMBER. CIRCUIT NUMBERS SHOWN ARE FOR REFERENCE ONLY. EC TO IDENTIFY AVAILABLE CIRCUITS FROM DEMO AND REUSE FOR NEW LAYOUT.
11. ALL PANEL LOCATIONS SHOWN ARE FOR CIRCUITING ONLY. FINAL LOCATIONS TO BE COORDINATED WITH ARCH DRAWINGS
12. NO. B/X ALLOWED WHATSOEVER. McCABLE, FLEXIBLE CONDUIT IS ALLOWED FOR FIXTURE WHIPS (NO MORE THAN 6'-0" IN LENGTH.)ELECTRICAL CONTRACTOR TO TYPE LABEL ALL PANELS. (NOT HAND WRITTEN.)
13. CONTRACTOR TO FURNISH A FILE SYSTEM FOR ALL EQUIPMENT AND MAINTENANCE OF EACH PIECE OF EQUIPMENT. ALL ELECTRICAL CIRCUITS HAVE TO BE MARKED AS TO WHAT EACH CONTROLS. CONTRACTOR TO FURNISH PHONE CONTACTS FOR WARRANTY SHOULD CONTRACTOR NOT BE LIABLE.
14. DWELLING UNIT RECEPTACLE SHALL BE TAMPER RESISTANCE IN ACCORDANCE WITH NATIONAL ELECTRICITY CODE (NEC).

SHEET NOTES

1. ALL ELECTRICAL PANEL, EQUIPMENTS, DISCONNECT AND METER INSTALL IN ACCORDANCE WITH NEC SECTION 110.26.

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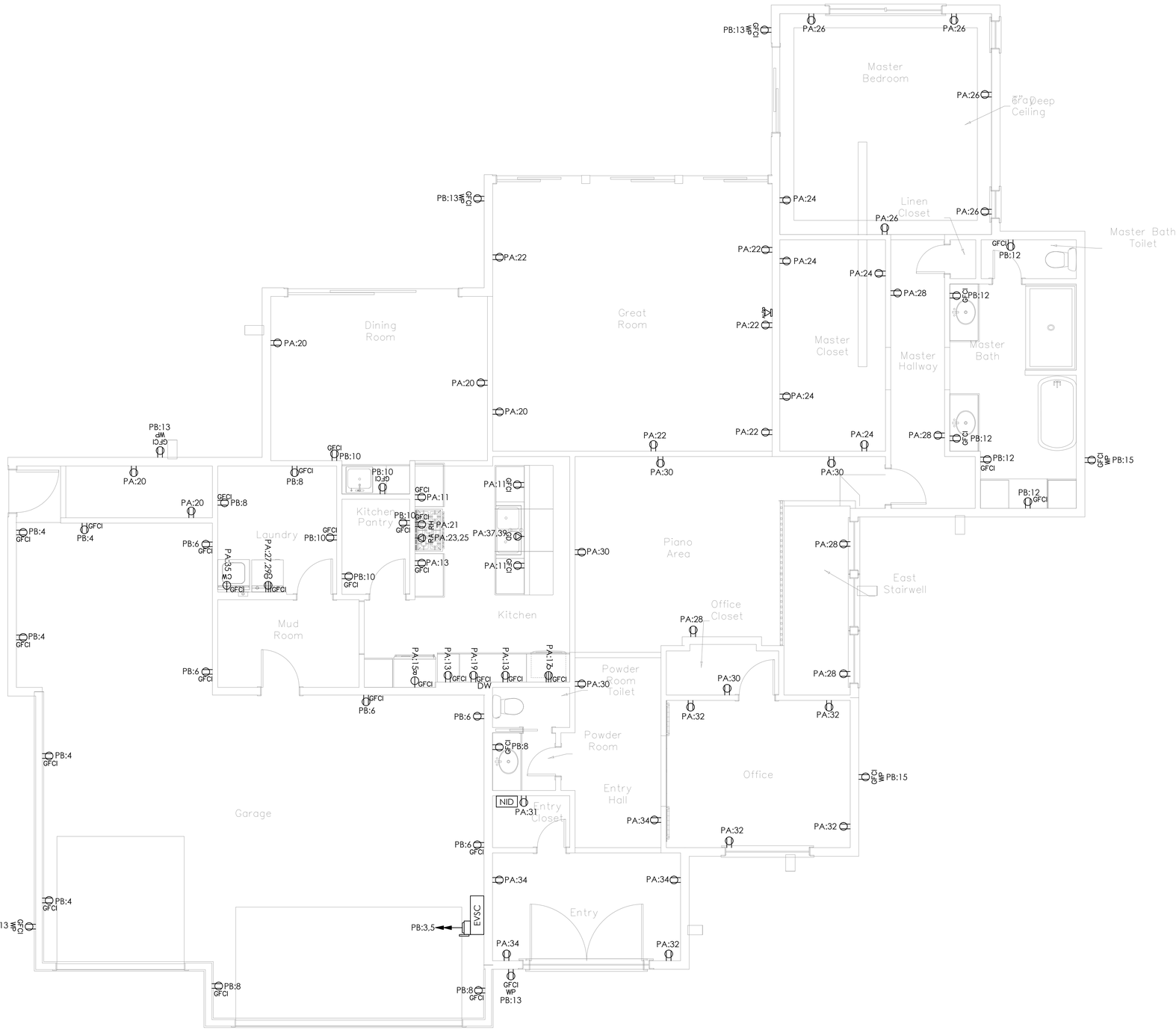
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ELECTRICAL FLOOR POWER PLAN

DRAWING NUMBER:

E-2.0



POWER NOTES

1. VERIFY EXACT REQUIREMENTS FOR ELECTRICAL EQUIPMENT WITH OWNER FOR THIS PROJECT. IN THE CASE OF A DISCREPANCY BETWEEN THE ACTUAL SELECTION OF THE EQUIPMENT AND THESE DRAWINGS, ADVISE THE OWNER BEFORE WORK
2. EXACT LOCATION OF RECEPTACLES & QUANTITY SHALL BE VERIFIED WITH ARCHITECT'S DOCUMENTS. POWER OUTLETS SHALL BE INSTALLED AT 16" A.F.F. TO BOTTOM OF OUTLET IN ALL PUBLIC AREAS. LIGHT SWITCHES SHALL BE INSTALLED AT 48" A.F.F. TO TOP OF SWITCHES IN ALL PUBLIC AREAS.
3. DRAWING IS DIAGRAMMATIC & INDICATES DIAGRAMMATIC WIRING & CIRCUIT DESIGNATIONS TO RECEPTACLES & DEVICES. FURNISH AND INSTALL ALL WIRING & CONDUITS AS REQUIRED FOR FULLY WORKABLE SYSTEM.
4. ELECTRICAL WHIPS/FLEXIBLE CONNECTIONS TO ELECTRIFIED FURNITURE SHALL BE A MAXIMUM LENGTH OF 18".
5. CONTRACTOR TO PROVIDE TYPED UPDATED PANEL DIRECTORY AS REQUIRED.
6. THE ELECTRICAL CONTRACTOR SHALL PROVIDE LABELING FOR ALL NEW PANELS AND SHALL RELABEL ALL EXISTING PANELS, AND CIRCUIT NUMBER.
7. CONTRACTOR SHALL PROVIDE AS-BUILT DRAWINGS INDICATING ACTUAL BRANCH CIRCUITING AND OUTLET LOCATION UPON COMPLETION OF WORK.
8. ELECTRICAL CONTRACTOR TO COORDINATE RECEPTACLE TYPE (NEMA TYPE) WITH EQUIPMENT (I.E. COPIER) SUPPLIER/MANUFACTURER PRIOR TO INSTALLATION.
9. PROVIDE 1¼"CONDUIT STUB-UPS AND BACK BOXES FOR ALL AV OUTLETS SHOWN. REFER TO AV PLANS FOR BOX SIZES.
10. ALL CIRCUITS TO BE TAGGED AT EACH END WITH APPROPRIATE CIRCUIT NUMBER. CIRCUIT NUMBERS SHOWN ARE FOR REFERENCE ONLY. EC TO IDENTIFY AVAILABLE CIRCUITS FROM DEMO AND REUSE FOR NEW LAYOUT.
11. ALL PANEL LOCATIONS SHOWN ARE FOR CIRCUITING ONLY. FINAL LOCATIONS TO BE COORDINATED WITH ARCH DRAWINGS
12. NO. B/X ALLOWED WHATSOEVER. McCABLE, FLEXIBLE CONDUIT IS ALLOWED FOR FIXTURE WHIPS (NO MORE THAN 6'-0" IN LENGTH.)ELECTRICAL CONTRACTOR TO TYPE LABEL ALL PANELS. (NOT HAND WRITTEN.)
13. CONTRACTOR TO FURNISH A FILE SYSTEM FOR ALL EQUIPMENT AND MAINTENANCE OF EACH PIECE OF EQUIPMENT. ALL ELECTRICAL CIRCUITS HAVE TO BE MARKED AS TO WHAT EACH CONTROLS. CONTRACTOR TO FURNISH PHONE CONTACTS FOR WARRANTY SHOULD CONTRACTOR NOT BE LIABLE.
14. DWELLING UNIT RECEPTACLE SHALL BE TAMPER RESISTANCE IN ACCORDANCE WITH NATIONAL ELECTRICITY CODE (NEC).

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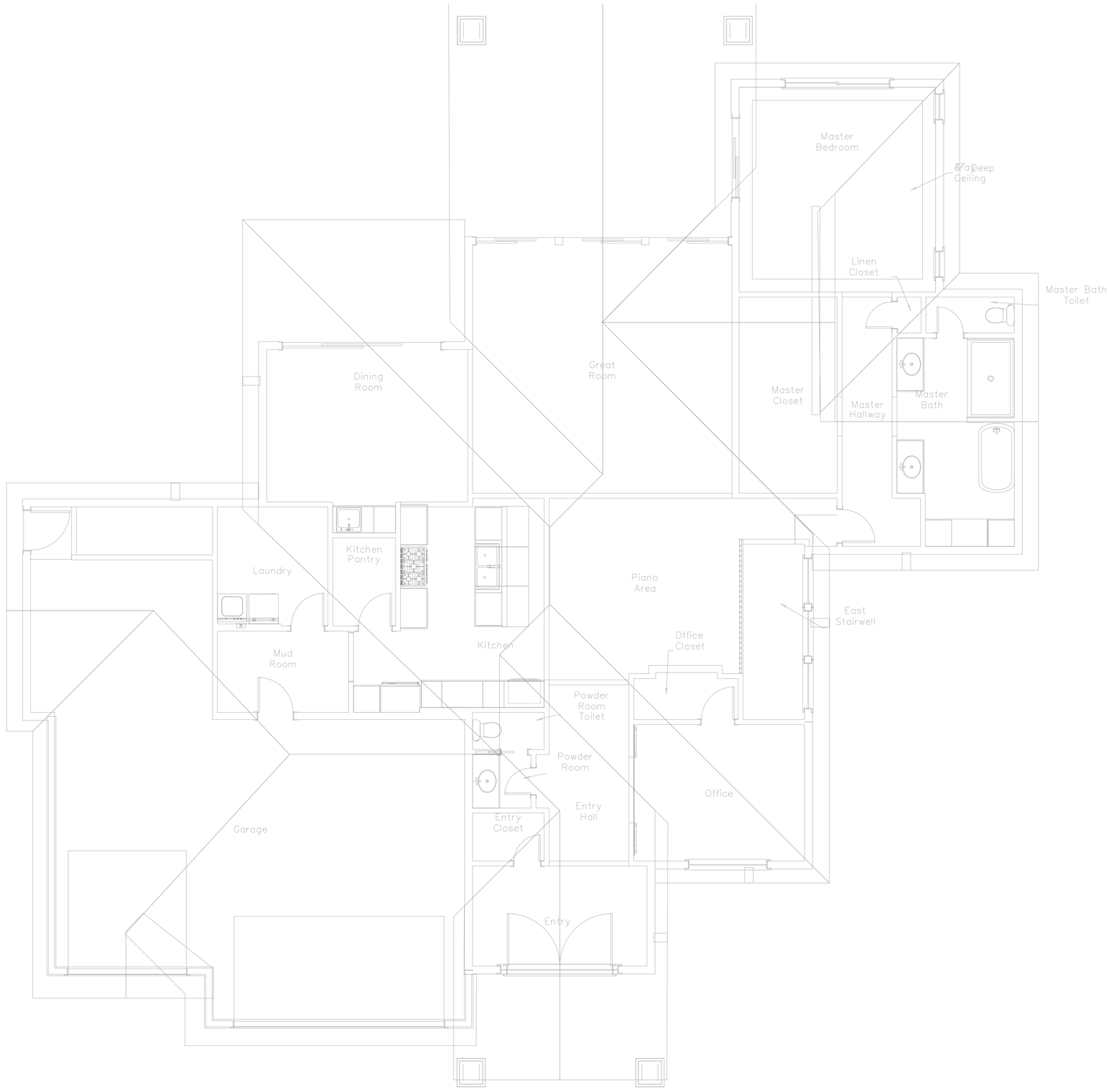
ELECTRICAL MAIN POWER PLAN

DRAWING NUMBER:

E-2.1

ELECTRICAL MAIN POWER PLAN

SCALE: 1/4"=1'=0"



ELECTRICAL ROOF POWER PLAN

SCALE: 1/4"=1'=0"

MAKMO DESIGN - DESIGN WITH A DIFFERENT APPROACH

ISSUE FOR:

FOR INTER REVIEW ONLY	
BID ONLY	
PERMITS SET	
CONSTRUCTION SET	

REVISIIONS:

NO.	DATE	DESCRIPTION

SEAL :

DATE:

PROJECT NUMBER	:
SCALE	:
DRAWN BY	:
CHECKED BY:	
SHEET TITLE	:

ELECTRICAL ROOF POWER PLAN

DRAWING NUMBER:

E-2.2

PROPOSED LOAD ANALYSIS @ 120/240V, 3PH, 4W, 60Hz

	LOAD DESCRIPTION	LOAD (VA)	DEMAND	TOTAL LOAD(VA)
1	LIGHTING DESIGN LOAD AS PER NEC 2020	6,000 S.F x 3.0VA		18000.00
2	CONSIDERING HIGHER INTERIOR LIGHTING LOAD	18000.00	125.00%	22500.00
3	CONSIDERING HIGHER EXTERIOR LIGHTING LOAD	200.00	125.00%	250.00
4	RECEPTACLE LOAD UPTO 10 KVA	10000.00	100.00%	10000.00
	RECEPTACLE LOAD ABOVE 10 KVA	4000.00	50.00%	2000.00
5	WASHER AND SMALL APPLIANCES	4500.00	100.00%	4500.00
6	FIXEDEND APPLIANCES - OVEN, DISPOSAL, DISHWASHER	3600.00	100.00%	3600.00
5	ELECTRICAL RANGE	8000.00	80.00%	6400.00
7	DRYER	5000.00	100.00%	5000.00
8	HVAC LOADS (AC, ACCU, EUH & INCLUDING WATER HEATER)	10000.00	100.00%	10000.00
	TOTAL DEMAND LOAD(VA)			64250
	TOTAL DEMAND CURRENT IN AMPS			**179**
	TOTAL DEMAND CURRENT @ 120/208 VOLTS WITH 10% SPARE (IN AMPS)			**196**

LOAD ANALYSIS

SCALE: N.T.S.

ELECTRICAL MATERIAL & EQUIPMENT

1. NO ELECTRICAL MATERIALS, APPARATUS, DEVICES, APPLIANCES, FIXTURES OR EQUIPMENT SHALL BE SOLD OR INSTALLED IN THE CITY, UNLESS THEY ARE IN CONFORMANCE WITH THE PROVISIONS OF THIS CODE, THE LAWS OF THE STATE OF COLORADO AND ANY APPLICABLE RULES AND REGULATIONS ISSUED UNDER THE AUTHORITY OF THE STATE STATUES.
2. THE MAKER'S NAME, TRADEMARK OR OTHER IDENTIFICATION SYMBOL SHALL BE PLACED ON ALL ELECTRICAL MATERIALS, APPARATUS, DEVICES, APPLIANCES, FIXTURES AND EQUIPMENT USED OR INSTALLED UNDER THE PROVISIONS OF THIS CODE.
3. ALL ELECTRICAL MATERIALS AND EQUIPMENT SHALL BE LISTED AND LABELED FOR THE INTENDED USE AND SHALL BE INCLUDED IN A LIST PUBLISHED BY AN APPROVED AGENCY.

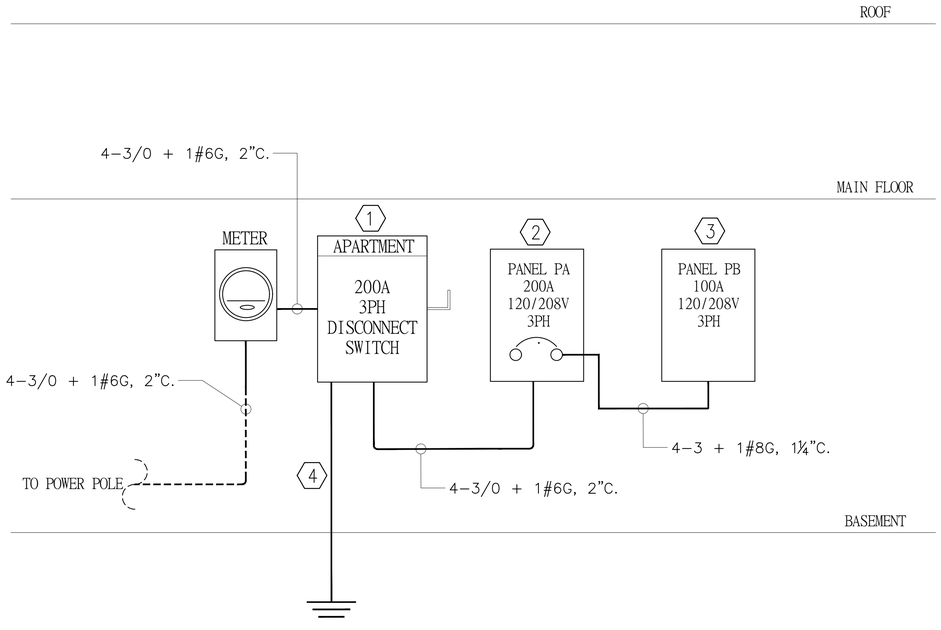
ELECTRICAL CODE SECTION 504.11 AVAILABLE FAULT CURRENT LABELING

IN LIEU OF THE MAXIMUM AVAILABLE FAULT CURRENT MARKING AS REQUIRED BY 110.24, A PERMANENTLY AFFIXED LABEL SHALL BE APPLIED WITH THE AVAILABLE FAULT CURRENT AT THE TIME OF INSTALLATION AND CALCULATION. THE LABEL SHALL BE 2" X 3" IN SIZE AND SHALL BE BLUE LETTERING ON A CONTRASTING BACKGROUND. THIS LABEL SHALL ALSO INCLUDE THE DATE OF THE CALCULATION

⬡ SINGLE LINE DIAGRAM NOTES

1. 200 AMP 3 POLE FUSED: 4 WIRE, 3R SWITCH WITH 3 (200) AMP FUSE.
2. 200 AMP 42 CRT. 3 PHASE WITH MAIN C/B PANEL.
3. 100 AMP 42 CRT. 3 PHASE SUB-PANEL.
4. 5/8 x 8'-0" COPPER GROUND ROD #4 WIRE - [GROUNDING AND INSULATION IN ACCORDANCE WITH NEC]

SINGLE LINE DIAGRAM:



ELECTRICAL RISER

SCALE: N.T.S

MAKMO DESIGN - DESIGN WITH A DIFFERENT APPROACH

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SEAL :

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CHECKED BY:	
SHEET TITLE	:

ELECTRICAL RISER & LOAD ANALYSIS

DRAWING NUMBER:

E-3.0

PANELBOARD

PANEL: **PA** **(NEW)**

208Y/120 VOLTS, **3** PHASE **4** WIRE

MCB **200A** INTERRUPTING RATING **22 KAIC** **MOUNTING:** SURFACE

NOTE:

CKT NO.	TRIP AMPS	DESCRIPTION OF LOAD	LOAD TYPE	LOAD (KVA)	MINIMUM BRANCH CIRCUIT	PER PHASE (KVA)			MINIMUM BRANCH CIRCUIT	LOAD (KVA)	LOAD TYPE	DESCRIPTION OF LOAD	TRIP AMPS	CKT NO.
						A	B	C						
1	20*	LIGHTING	L	0.80	2-12 + 1#12G, 3/4"C.	1.70			2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	2
3	20*	LIGHTING	L	0.75	2-12 + 1#12G, 3/4"C.		1.65		2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	4
5	20*	LIGHTING	L	0.85	2-12 + 1#12G, 3/4"C.			1.75	2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	6
7	20*	CEILING FAN	L	0.70	2-12 + 1#12G, 3/4"C.	1.60			2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	8
9	20*	EXTERNAL LIGHTING	L	0.40	2-12 + 1#12G, 3/4"C.		1.30		2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	10
11	20*	SMALL APPLIANCES	K	1.50	2-12 + 1#12G, 3/4"C.			2.40	2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	12
13	20*	SMALL APPLIANCES	K	1.50	2-12 + 1#12G, 3/4"C.	2.40			2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	14
15	20	REFRIGERATOR	K	1.20	2-12 + 1#12G, 3/4"C.		2.10		2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	16
17	20	OVEN	K	1.30	2-12 + 1#12G, 3/4"C.			2.20	2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	18
19	20	DISHWASHER	K	1.30	2-12 + 1#12G, 3/4"C.	2.20			2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	20
21	20	RANGE HOOD	K	0.36	2-12 + 1#12G, 3/4"C.		1.26		2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	22
23	50/2P	RENGE	K	4.00	2-6 + 1#10G, 3/4"C.			4.90	2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	24
25			K	4.00		4.90			2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	26
27	30/2P	DRYER	E	2.50	2-10 + 1#10G, 3/4"C.		3.40		2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	28
29			E	2.50				3.40	2-12 + 1#12G, 3/4"C.	0.90	R	RECEPTACLE	20*	30
31	20*	NID	E	0.18	2-12 + 1#12G, 3/4"C.	0.54			2-12 + 1#12G, 3/4"C.	0.36	R	RECEPTACLE	20*	32
33	20*	S/CO DETECTOR	E	0.30	2-12 + 1#12G, 3/4"C.		0.66		2-12 + 1#12G, 3/4"C.	0.36	R	RECEPTACLE	20*	34
35	20	WASHER	E	1.50	2-12 + 1#12G, 3/4"C.			1.50				SPARE	20	36
37	30/2P	GARBAGE DISPOSAL	K	1.00	2-10 + 1#10G, 3/4"C.	11.00				10.00	M			38
39			K	1.00			11.00		4-3 + 1#8G, 11/4"C.	10.00	M	PANEL PB	100/3P	40
41	20	SPARE						10.00		10.00	M			42
						24.34	**21.37**	**26.15**						

CLASSIFICATION	CONNECTED LOAD (KVA)				DEMAND			DEMAND LOAD (KVA)			PANEL TOTAL LOAD		
LIGHTING	L	3.50			100%			3.50					
RECEPTACLE	R	14.22			100%			14.22			**TOTAL CONNECTED LOAD**	71.86	KVA
HVAC	H	0.00			100%			0.00			**TOTAL DEMAND LOAD**	56.85	KVA
MISCELLANEOUS	M	30.00			70%			21.00			**TOTAL CONNECTED CURRENT**	199.70	AMP
EQUIPMENTS	E	6.98			100%			6.98			**TOTAL DEMAND CURRENT**	158.00	AMP
KITCHEN LOAD	K	17.16			65%			11.15			**CIRCUIT BREAKER RATING**	165.90	AMP

PANEL: **PB** **(NEW)**

208Y/120 VOLTS, **3** PHASE **4** WIRE

MLO **100A** INTERRUPTING RATING **22 KAIC** **MOUNTING:** SURFACE

NOTE:

CKT NO.	TRIP AMPS	DESCRIPTION OF LOAD	LOAD TYPE	LOAD (KVA)	MINIMUM BRANCH CIRCUIT	PER PHASE (KVA)			MINIMUM BRANCH CIRCUIT	LOAD (KVA)	LOAD TYPE	DESCRIPTION OF LOAD	TRIP AMPS	CKT NO.
						A	B	C						
1	20	CAT5/TV	R	0.90	2-12 + 1#12G, 3/4"C.	1.80			2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	2
3	30/2P	EVSC (ELECTRIC VEHICLE	E	2.50	2-10 + 1#10G, 3/4"C.		3.40		2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	4
5		CHARGING STATION)	E	2.50				3.40	2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	6
7	20	GAS WH-1	H	0.36	2-12 + 1#12G, 3/4"C.	1.26			2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	8
9	20	CIRCULATION PUMP CP-1	H	0.10	2-12 + 1#12G, 3/4"C.		1.00		2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	10
11	20	WP/GFI RECEPTACLE	R	0.90	2-12 + 1#12G, 3/4"C.			1.80	2-12 + 1#12G, 3/4"C.	0.90	R	GFI RECEPTACLE	20	12
13	20	WP/GFI RECEPTACLE	R	0.90	2-12 + 1#12G, 3/4"C.	0.90						SPARE	20	14
15	20	WP/GFI RECEPTACLE	R	0.36	2-12 + 1#12G, 3/4"C.		0.36					SPARE	20	16
17	20	SPARE						0.00				SPARE	20	18
19	20	SPARE				0.00						SPARE	20	20
21	20	SPARE					0.00					SPARE	20	22
23	20	SPARE						0.00				SPARE	20	24
25	20	SPARE				0.00						SPARE	20	26
27	20	SPARE					0.00					SPARE	20	28
29	20	SPARE						0.00				SPARE	20	30
31	20	SPARE				0.00						SPARE	20	32
33	20	SPARE					0.00					SPARE	20	34
35	20	SPARE						0.00				SPARE	20	36
27	20	SPARE				0.00						SPARE	20	38
29	20	SPARE					0.00					SPARE	20	40
41	20	SPARE						0.00				SPARE	20	42
						3.96	**4.76**	**5.20**						

CLASSIFICATION	CONNECTED LOAD (KVA)				DEMAND			DEMAND LOAD (KVA)			PANEL TOTAL LOAD		
LIGHTING	L	0.00			100%			0.00					
RECEPTACLE	R	8.46			100%			8.46			**TOTAL CONNECTED LOAD**	13.92	KVA
HVAC	H	0.46			100%			0.46			**TOTAL DEMAND LOAD**	13.92	KVA
MISCELLANEOUS	M	0.00			70%			0.00			**TOTAL CONNECTED CURRENT**	38.68	AMP
EQUIPMENTS	E	5.00			100%			5.00			**TOTAL DEMAND CURRENT**	38.68	AMP
KITCHEN LOAD	K	0.00			65%			0.00			**CIRCUIT BREAKER RATING**	40.62	AMP

PANEL SCHEDULE NOTES:

1. '*' DENOTES AFCI TYPE CIRCUIT BREAKER.

ELECTRICAL PANEL SCHEDULES

SCALE: N.T.S

ISSUE FOR:

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PROJECT NUMBER	:
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ELECTRICAL PANEL SCHEDULE

DRAWING NUMBER:

E-4.0